BINGO

Virous-24

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Introduction

The Bingo Game Project is a Python bingo game project designed to be played against the computer using the Tkinter GUI library. The game allows players to mark numbers on a grid, with the goal of completing a row, column, or diagonal to win.

why we selects this problem?

Key features of the project include:

01 Easy to use interface: Simple graphical user interface for easy gameplay

02 Multiple difficulty levels: Different levels of challenge.

03 Helps us apply what we have learned in Python

Analysis and Design

Main Functions:

create_board(size): Creates a random Bingo board of a given size.
mark_board(board, number): Marks the number on the board.

count_bingos(board): Counts the number of completed Bingo lines.
get_computer_move(board, valid_numbers, level): Determines the computer's move based on difficulty.

update_gui_board(frame, board): Updates the game board in the GUI.
save_game_to_file( ): Saves the current game state.
load_game_from_file( ): Loads a saved game.

delete_saved_game( ): Deletes the saved game.
handle_escape( ): Handles the "Escape" key (save, exit, or continue).
declare_winner( ): Declares the winner of the game

Explanation of the theory behind our code:

01

Algorithm Overview:

- Create or load the board.
- The player and computer take turns, marking numbers on their boards.
- After each move, check for Bingo.
- The game ends when a player reaches 5 Bingos.

02

The code is based on programming techniques such as:

- Generate random numbers using the random library.
- Interact with the user via the Tkinter interface.
- Implement loops and lists to check the winning condition.

Test Cases:

Create Board:

•Input: create_board(5)

•Output: A 5x5 board with random numbers from 1 to 25

Mark Number on Board:

•Input: mark_board(user_board, 5)

•Output: The number 5 is replaced with "X" on the board.

Count Bingos:

•Input: A board with a filled row of "X".

•Output: count_bingos(user_board) returns 1.

Player and Computer Turns Logic:

Input: Player chooses number 7.

Output: Board is updated with player's move, then computer's move

Win Condition 5 Bingos:

Input: Player or computer achieves 5 Bingos.

Output: Winner is announced, and the game ends

Save Game on "Escape" Key:

Input: Pressing the "Escape" key.

Output: Popup window to save, exit, or continue.

Challenges and problems:

01

User Experience:

Make the interface intuitive and easy to use.

Solution: Design a simple and clear interface.

02

User input:

Validating input can be difficult.

Solution: Implement input validation and error messages.

03

Performance with large boards:

Performance issues.

Solution: Improve logic to increase efficiency.

Conclusion and Discussion

A bingo game was successfully developed with different difficulty levels for AI, providing an enjoyable user experience. The project implemented the basics of game development such as handling user inputs, managing game state, and using the computer for the opponent. The graphical user interface (GUI) also enhanced the interaction with the game.

Suggestions for improvement:

01 Develop a portable version.

02 Add sound effects and animations.

03 Add a two-player mode

References:

Python Documentation: https://docs.python.org

Tkinter Documentation: https://docs.python.org/3/library/tkinter.html

Future work:

Implement an online multiplayer mode.

Thanks!

any questions?